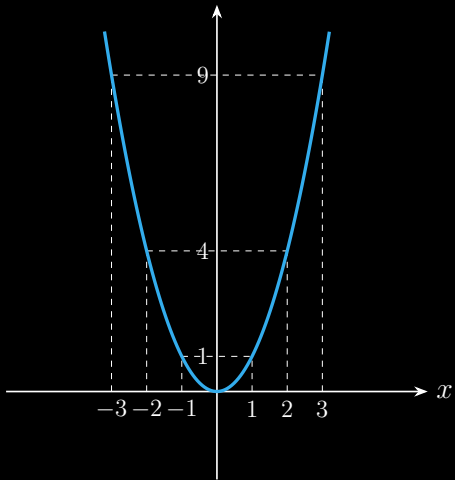
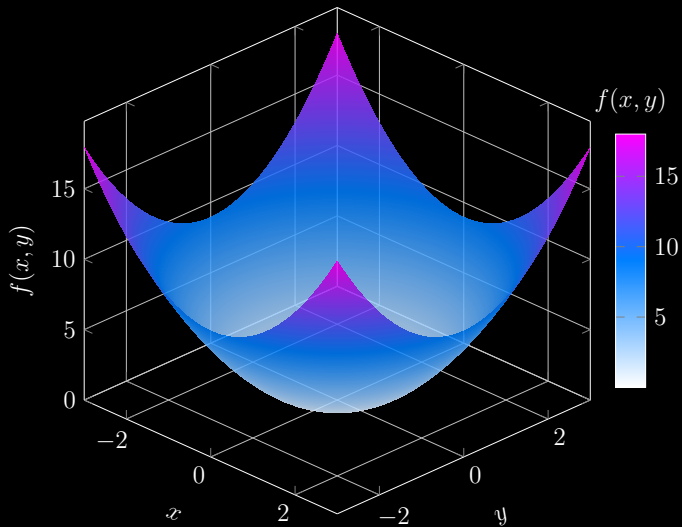


Quadratic Function

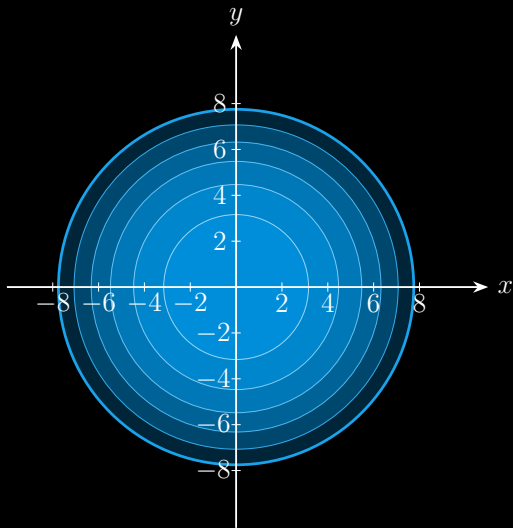
$$f(x) = x^2$$



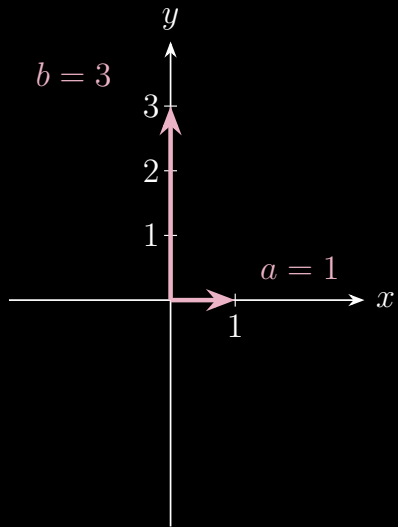
Quadratic Function $f(x, y) = x^2 + y^2$

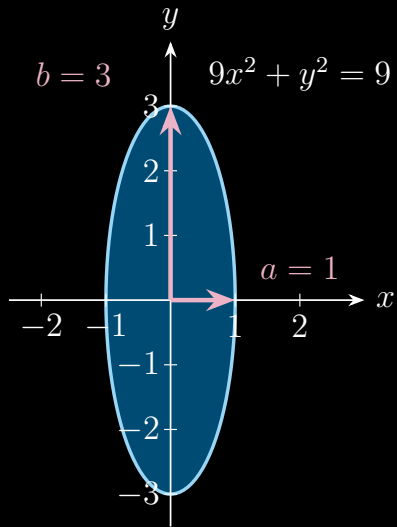


Quadratic Function $f(x, y) = x^2 + y^2$

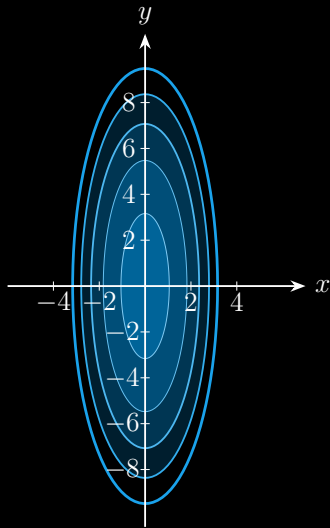


Ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

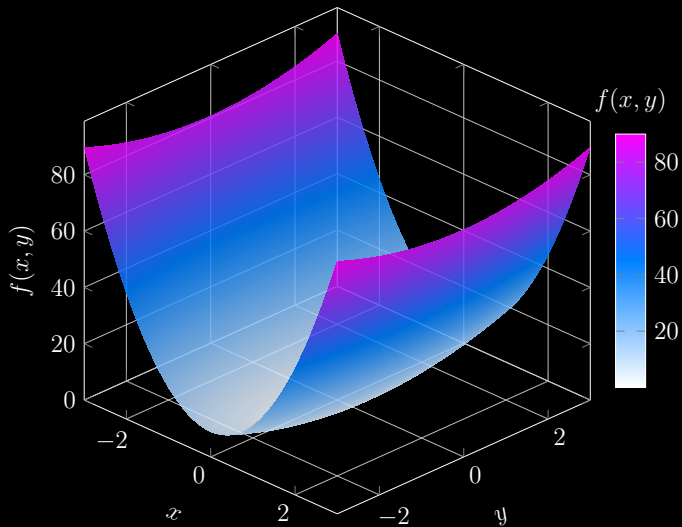




Quadratic Function $f(x, y) = 9x^2 + y^2$



Quadratic Function $f(x, y) = 9x^2 + y^2$



Quadratic Function

$$f(x, y) = 9x^2 + 4xy + y^2 = [x \ y] \underbrace{\begin{bmatrix} 9 & 2 \\ 2 & 1 \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} x \\ y \end{bmatrix}$$

Eigenvalue decomposition:

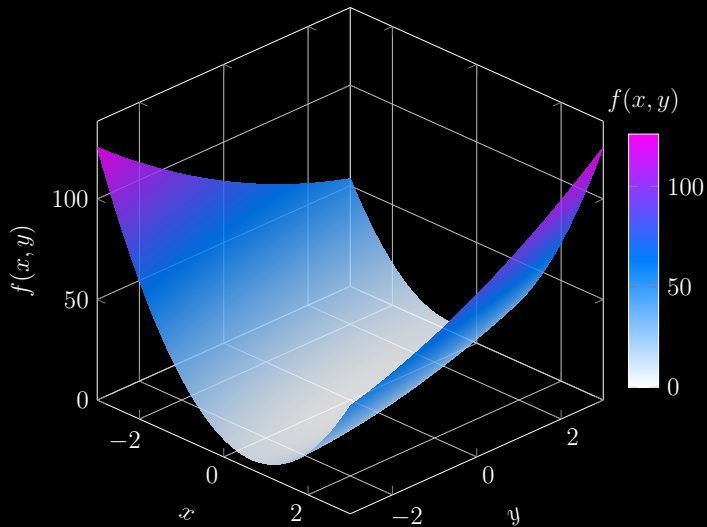
$$\mathbf{A} = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top$$

with eigenvectors:

$$\mathbf{u}_1 = \begin{bmatrix} 0.973 \\ 0.230 \end{bmatrix} \quad \mathbf{u}_2 = \begin{bmatrix} -0.230 \\ 0.973 \end{bmatrix}$$

```
1 import numpy as np
2
3 A = np.array([[9, 2], [2, 1]])
4 eig_val, eig_vec = np.linalg.eig(A)
5
6 print('lambda_1 = ', eig_val[0])
7 print('lambda_2 = ', eig_val[1])
8 print('u_1 = ', eig_vec[:, 0])
9 print('u_2 = ', eig_vec[:, 1])
```


Quadratic Function $f(x, y) = 9x^2 + 4xy + y^2$



Thanks for your attention!

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